

COLMAR-MEYENHEIM AB, France

WMO: 071970

Lat: 47.9286N

Lon: 7.4078E

Elev: 693

StdP: 14.33

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.8	(c) 20.8	(d) 12.7	(e) 10.8	(f) 19.5	(g) 16.6	(h) 13.1	(i) 24.3	(j) 29.1	(k) 48.5	(l) 25.6	(m) 46.6	(n) 4.2	(o) 20	(p) 0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	90.6	69.8	86.4	68.3	83.3	67.4	71.5	86.1	70.0	83.0	68.6	80.5	7.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.2	102.4	76.4	65.7	97.3	74.7	64.4	93.0	73.4	35.6	86.3	34.3	83.0	33.2	80.7	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.8	(b) 19.8	(c) 17.3	(e) 11.2	(f) 95.2	(g) 4.4	(h) 4.0	(i) 8.0	(j) 98.0	(k) 5.4	(l) 100.4	(m) 3.0	(n) 102.6	(o) -0.3	(p) 105.5		
(a) 22.8	(b) 19.8	(c) 17.3	(e) 10.6	(f) 74.4	(g) 4.4	(h) 1.6	(i) 7.5	(j) 75.6	(k) 4.9	(l) 76.5	(m) 2.4	(n) 77.4	(o) -0.7	(p) 78.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	35.9	38.7	44.7	51.3	59.6	65.7	68.7	68.0	60.1	52.7	42.9	37.9
	DBStd	13.51	8.66	7.57	7.01	6.73	6.34	6.47	5.63	5.86	5.86	6.77	7.40	7.81
	HDD50	1687	439	322	192	65	5	0	0	0	2	51	231	380
	HDD65	5056	902	737	630	411	189	71	27	34	169	382	663	841
	CDD50	2514	2	5	27	105	304	470	579	558	305	136	18	5
	CDD65	407	0	0	0	1	23	91	141	127	22	2	0	0
	CDH74	4417	0	0	0	52	329	979	1461	1307	262	27	0	0
	CDH80	1528	0	0	0	4	79	328	544	510	61	2	0	0
Wind	WSAvg	7.2	8.3	8.4	8.7	7.6	7.2	6.3	6.2	5.5	6.1	6.5	7.4	8.1
Precipitation	PrecAvg	28.2	1.7	1.7	1.7	2.0	3.1	2.8	2.8	2.8	2.4	2.7	2.2	2.3
	PrecMax	37.2	4.6	4.2	6.4	4.7	6.1	5.7	6.2	6.2	5.8	7.7	5.7	5.0
	PrecMin	18.6	0.4	0.1	0.3	0.3	1.2	1.2	1.1	0.5	0.6	0.1	0.0	0.0
	PrecStd	4.8	1.1	1.0	1.3	1.2	1.4	1.2	1.2	1.3	1.2	1.4	1.3	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.8	63.3	71.2	79.7	86.6	92.9	96.1	96.2	86.2	78.2	65.3	57.4
		MCWB	49.9	50.2	54.4	59.7	66.6	71.1	70.9	70.5	65.7	62.5	55.7	51.2
	2%	DB	53.7	57.9	65.5	74.1	81.6	87.8	90.3	90.2	80.8	71.7	60.0	54.6
		MCWB	49.1	48.4	52.2	57.1	64.0	68.5	70.1	69.6	66.0	60.8	53.4	50.0
	5%	DB	50.9	53.9	60.4	69.3	77.5	83.7	85.5	85.7	76.6	67.6	56.5	52.0
		MCWB	46.8	47.3	50.2	54.9	62.3	67.6	68.5	68.5	63.7	59.1	51.6	48.4
	10%	DB	47.6	50.4	56.5	65.1	73.3	79.8	82.1	81.6	72.3	63.6	53.5	49.4
		MCWB	44.3	45.5	48.6	53.4	60.0	66.1	67.6	67.5	61.6	57.2	49.6	46.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	51.8	56.1	61.2	68.8	73.8	73.7	72.9	69.2	64.5	57.8	53.0
		MCDB	55.0	59.1	67.0	76.9	82.4	88.2	89.3	87.7	80.5	73.9	62.8	55.9
	2%	WB	49.7	49.8	53.7	58.2	66.0	70.6	71.5	71.1	66.6	61.9	54.6	50.8
		MCDB	53.3	56.3	61.6	70.7	79.0	83.7	86.2	85.8	77.7	69.0	58.7	53.7
	5%	WB	46.9	47.8	51.5	56.2	63.5	68.8	70.0	69.6	64.7	60.0	52.1	48.7
		MCDB	50.3	53.5	58.4	67.4	74.1	80.6	82.9	82.5	74.0	66.0	55.6	51.6
	10%	WB	44.6	45.7	49.5	54.2	61.3	67.0	68.4	68.2	62.7	58.0	50.1	46.5
		MCDB	47.3	50.1	55.9	63.3	70.5	77.7	79.8	79.6	70.2	63.0	53.2	49.3
Mean Daily Temperature Range	5% DB	MDBR	11.1	14.8	18.2	21.0	20.2	20.9	21.4	21.7	20.1	17.4	12.9	10.7
		MCDBR	14.1	21.9	26.3	29.9	28.2	28.3	28.5	29.6	27.8	23.9	17.5	13.3
		MCWBR	10.3	13.8	15.2	14.9	12.9	12.3	11.3	11.6	13.3	13.7	11.7	9.8
	5% WB	MCDBR	13.1	19.1	21.7	26.4	25.1	25.8	25.7	26.3	23.5	21.3	16.2	11.9
MCWBR		10.2	12.6	13.1	13.3	12.0	11.6	10.8	11.2	11.8	12.8	11.4	9.3	
Clear-Sky Solar Irradiance	taub	0.314	0.338	0.370	0.406	0.415	0.421	0.418	0.413	0.389	0.367	0.334	0.310	
	taud	2.450	2.380	2.315	2.234	2.254	2.274	2.274	2.305	2.350	2.398	2.446	2.449	
	Ebn at Noon	242	259	267	268	270	268	266	263	258	244	231	227	
	Edh at Noon	21	28	34	41	42	41	41	38	33	27	21	19	
All-Sky Solar Radiation	RadAvg	320	569	961	1370	1606	1876	1771	1538	1169	716	369	258	
	RadStd	46	95	150	199	168	166	165	151	115	62	42	49	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (101 neighbors)		N/A	N/A	N/A	N/A	+0.68	+0.52	N/A	N/A	N/A	+29

Nomenclature: See separate page